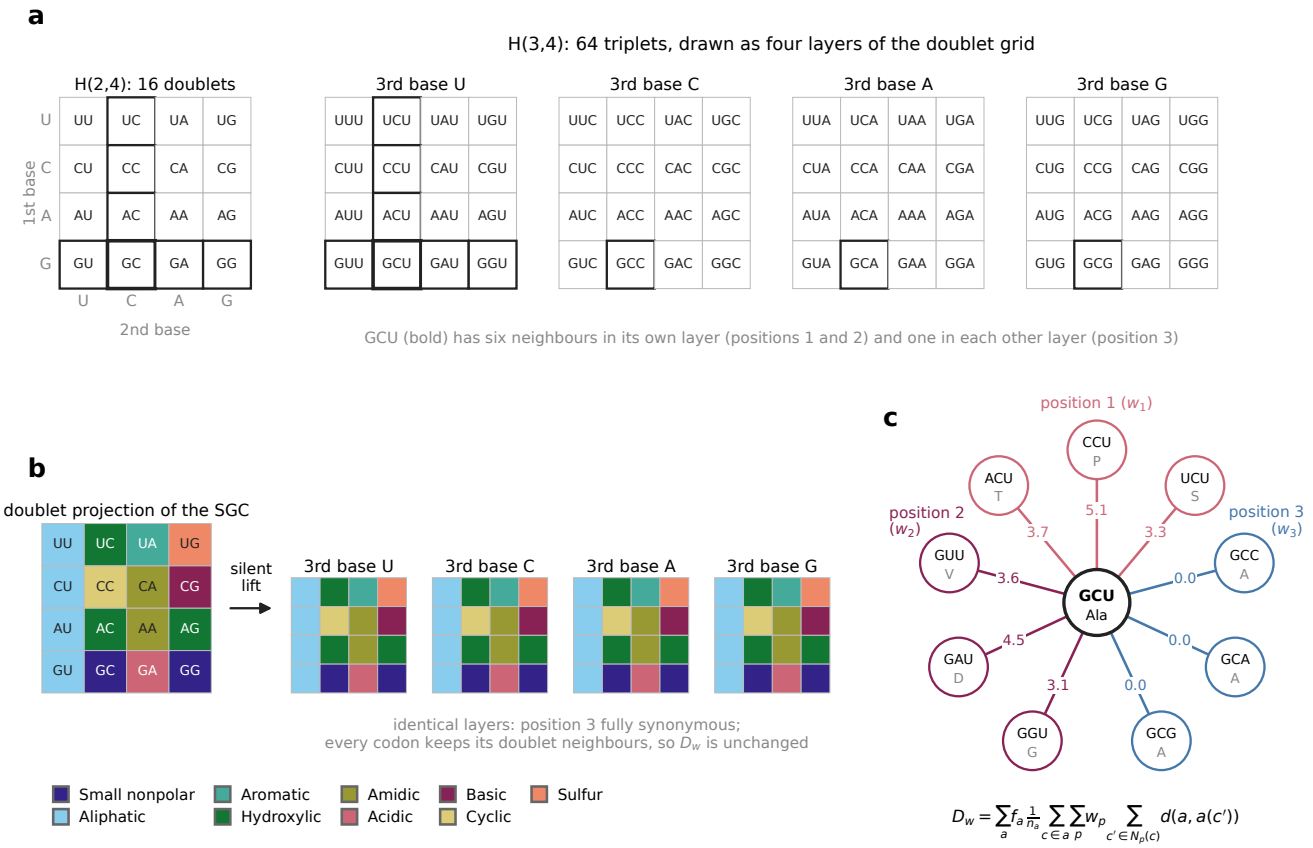


Figure 1. Codes on the Hamming graph and the silent lift. (a) $H(2, 4)$ drawn as a 4 × 4 grid (rows, first base; columns, second base), and $H(3, 4)$ drawn as four copies of that grid, one per third base (after Lenstra 2015). Two codons are adjacent when they differ at one position: GCU (bold outline) has six neighbours in its own layer, at positions 1 and 2, and one neighbour in each of the other three layers, at position 3. (b) The doublet projection of the SGC (left; nine classes, coloured by class) and its silent lift (right): every layer of the triplet code is a copy of the doublet code, so position 3 is fully synonymous and every codon keeps the non-synonymous neighbours it had as a doublet. (c) The neighbourhood of GCU (Ala) in the SGC, with the physicochemical distance d to each neighbouring amino acid; position-3 neighbours are synonymous ($d = 0$). The per-residue cost D_W sums these distances over all neighbours, weighting each position by its misreading rate w_p and each amino acid by its usage f_a.

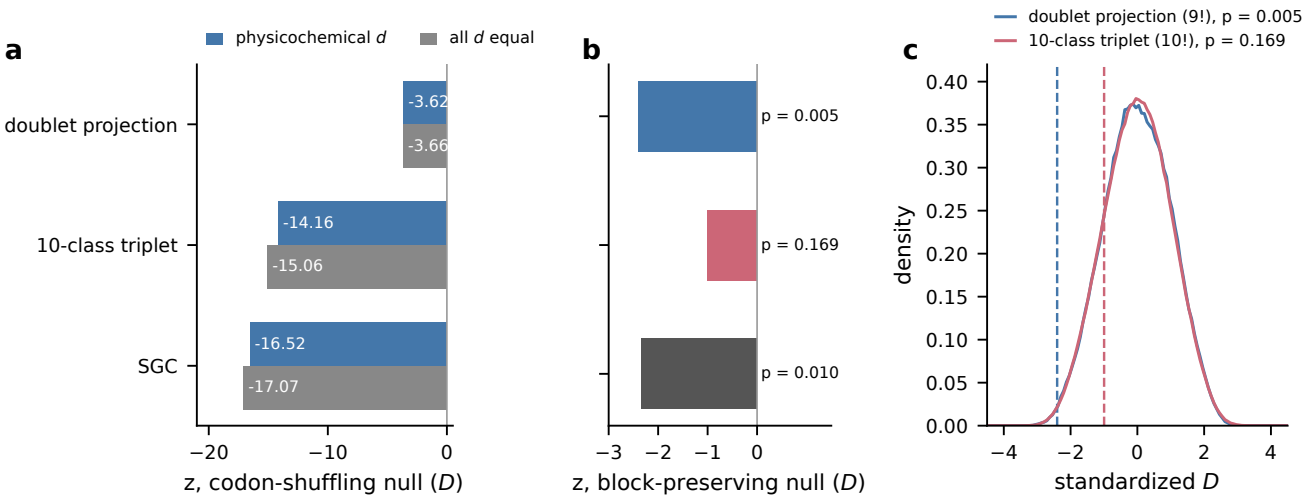


Figure 2. Codon-shuffling nulls measure synonym clustering. (a) z-scores of the noise distortion D of the doublet projection, the ten-class triplet projection and the SGC against the codon-shuffling null (10^6 shuffles, seed 20260924), with physicochemical distances (blue) and with every pair of amino acids at the same distance (grey). (b) The same codes against the block-preserving null; the two projections were enumerated exhaustively (9! and 10! codes) and the SGC was sampled 10^7 times. p is the fraction of codes strictly better than the observed code. (c) Exhaustive block-preserving null distributions of D for the two projections, standardized to zero mean and unit variance; dashed lines mark the observed codes.

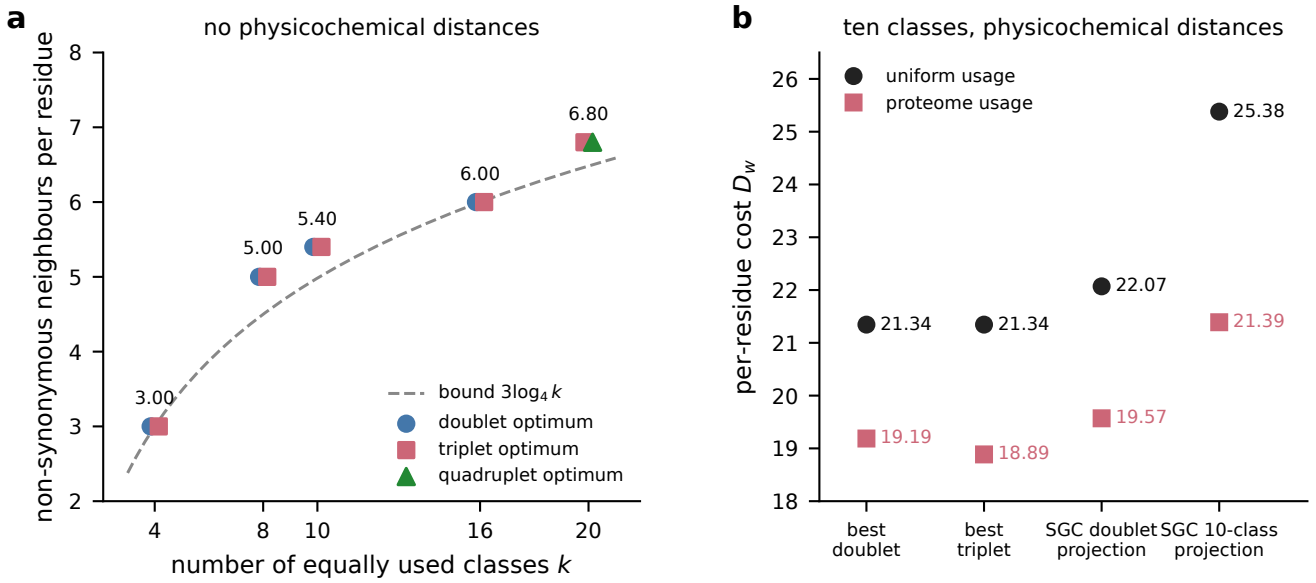


Figure 3. Codon length does not change the achievable cost. (a) Minimum number of non-synonymous neighbours per residue, without physicochemical distances, for doublets (circles) and triplets (squares) with k equally used classes, and for triplets and quadruplets with 20 classes (k = 20). The dashed line is the edge-isoperimetric bound 3 $\log_4 k$, attained at k = 4 and 16. (b) Minimum per-residue cost D_W for ten classes with physicochemical distances under uniform (circles) and proteome (squares) usage, for the best doublet and triplet codes and for the two projections of the SGC.

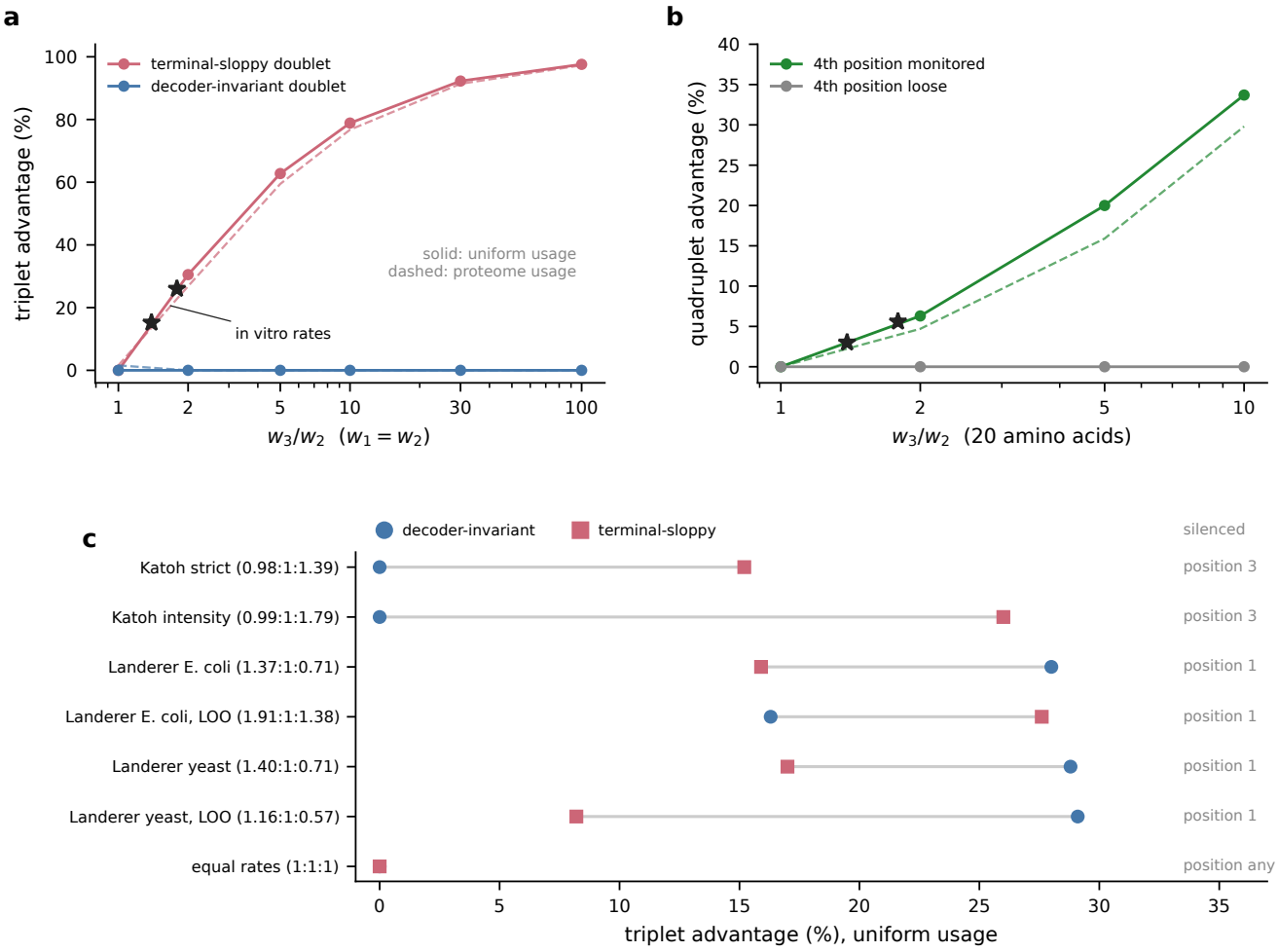


Figure 4. The advantage of a longer code is the cost of the channel it discards. (a) Advantage of the best triplet code over the best doublet code, $1 - D_3^*/D_2^*$, against w_3/w_2 (with $w_1 = w_2$), under the terminal-sloppy and decoder-invariant assumptions; solid lines, uniform usage; dashed lines, proteome usage. Stars mark the in vitro misreading rates of Katoh and Suga (2023). (b) Advantage of the best quadruplet code over the best triplet code for 20 amino acids, when the fourth position is read as accurately as position 2 (monitored) or at the position-3 rate (loose); stars mark the in vitro rates. (c) Triplet advantage at the seven published rate sets of table 2 under the two assumptions about doublet decoding (uniform usage), with the position the best triplet code silences.